

AI Product/Engineer Intern Assignment

Project Title

AI-Powered Sports Quiz Generation Agent

Objective

Develop an AI agent capable of generating engaging sports-related multiple-choice quizzes for social media. The system should leverage Retrieval-Augmented Generation (RAG) using a vector database and web search to ensure factual accuracy and fresh content.

Problem Statement

Traditional sports content on social media is largely limited to news, highlights, and opinion-based posts. The goal of this project is to introduce a new interactive content format by automatically generating sports quizzes that encourage audience participation and increase engagement.

The AI agent should create unique, factually accurate quizzes based on the selected sport and difficulty level, making it easier for content creators to publish interactive posts consistently.

Functional Requirements

- Select a sport (e.g., Cricket, Football, Tennis, Badminton, Basketball, etc.).
 - Choose a difficulty level (Easy, Medium, Hard).
 - Generate multiple-choice sports quizzes.
 - Regenerate quizzes when requested.
 - Produce factually accurate questions using retrieved knowledge.
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AI Agent Requirements

- Use web search to gather recent sports information when required.
 - Retrieve relevant sports knowledge from a ChromaDB vector database.
 - Combine retrieved context with the LLM to generate high-quality quiz questions.
 - Avoid hallucinations by grounding responses using retrieved information.
 - Generate fresh and diverse quiz questions for every request.
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Expected Output

- Sport Name
 - Difficulty Level
 - Four to Five quiz question
 - Four answer options (A, B, C, D)
 - Correct answer
 - Short explanation of the answer
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Example Output

Sport: Badminton

Difficulty: Medium

Question:

Which country won the Thomas Cup in 2022?

A. Indonesia

B. India

C. China

D. Denmark

Correct Answer: B. India

Explanation:

India won its first-ever Thomas Cup title in 2022 after defeating Indonesia in the Thomas Cup final.

Expected Deliverables

- Working AI-powered quiz generation dashboard
- Integration with web search
- ChromaDB-based knowledge retrieval
- Clean and well-documented code
- README with setup instructions and project overview

Evaluation Criteria

- Accuracy of generated quizzes
- Quality and relevance of retrieved information
- Effective use of the AI agent
- Correct implementation of ChromaDB knowledge retrieval
- User experience and usability
- Code quality and documentation